Project Design Phase-II

Technology Stack (Architecture & Stack)

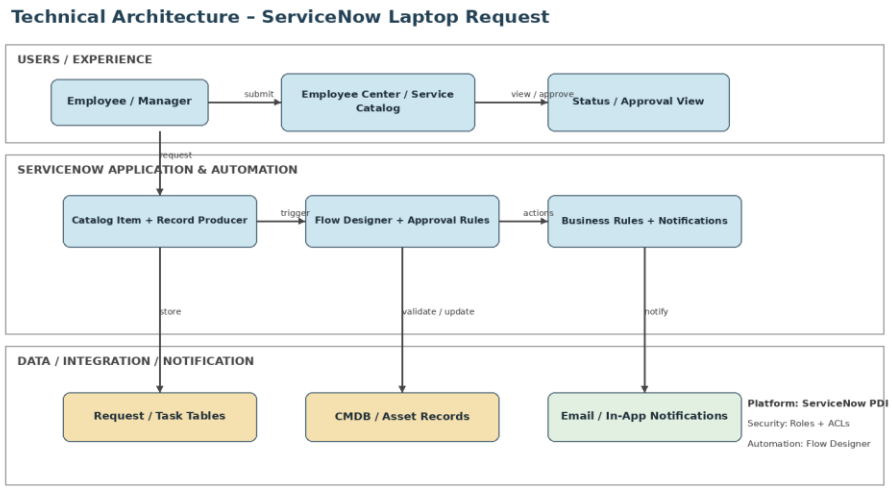
Date	10 September 2026
Team ID	SWUID2026190374
Project Name	ServiceNow-Based Employee Laptop Request Application
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

ServiceNow Employee Laptop Request – Technical Architecture

Reference: ServiceNow platform architecture, Service Catalog and Flow Designer



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Employee request and manager approval experience	ServiceNow Employee Center / Service Catalog
2	Application Logic-1	Create and validate request	Catalog Item / Record Producer
3	Application Logic-2	Route approval and create tasks	Flow Designer
4	Application Logic-3	State logic, validation and notifications	Business Rules / Notifications
5	Database	Requests, tasks, users and workflow data	ServiceNow platform tables / GlideRecord
6	Cloud Database	Managed platform data storage	ServiceNow Cloud Platform
7	File Storage	Request attachments when required	ServiceNow Attachments
8	External API-1	Optional HR or identity integration	REST APIs / IntegrationHub
9	External API-2	Optional asset/inventory integration	CMDB / REST
10	Machine Learning Model	Not required for core workflow	N/A
11	Infrastructure (Server / Cloud)	Hosted application instance	ServiceNow PDI / Cloud Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No external framework required for the core PDI application.	ServiceNow platform
2	Security Implementations	Role-based access, ACLs and authenticated sessions.	ServiceNow Roles / ACLs
3	Scalable Architecture	Catalog, workflow, data and integrations are modular and extensible.	Catalog + Flow Designer

S.No	Characteristics	Description	Technology
4	Availability	Uses the managed ServiceNow cloud platform.	ServiceNow Cloud
5	Performance	Efficient queries, server-side automation and asynchronous notifications.	Glide APIs / Flow Designer

References:

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d